Git & Github Basic Commands

```
sharafkamaal@192 devops-techiehorizon % git --version
git version 2.50.1 (Apple Git-155)
sharafkamaal@192 devops-techiehorizon % touch 123
sharafkamaal@192 devops-techiehorizon % git status
fatal: not a git repository (or any of the parent directories): .git
sharafkamaal@192 devops-techiehorizon % git init
Initialized empty Git repository in /Users/sharafkamaal/devops-techiehorizon/.git/
sharafkamaal@192 devops-techiehorizon % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .DS_Store
        123
        VirtualBox-7.2.20-175154-macOSArm64.dmg
        aws.pem
        devops-pdf/
        devops-video/
        ha-proxy-server.pem
        if-else
        kamaal/
        server-1.pem
        server-2.pem
        ubuntu-26.04.1-desktop-arm64.iso

nothing added to commit but untracked files present (use "git add" to track)
```

1. git init – Initializes a new Git repository in the current directory.

2. git status – Shows the current state of files and changes in the Git repository.

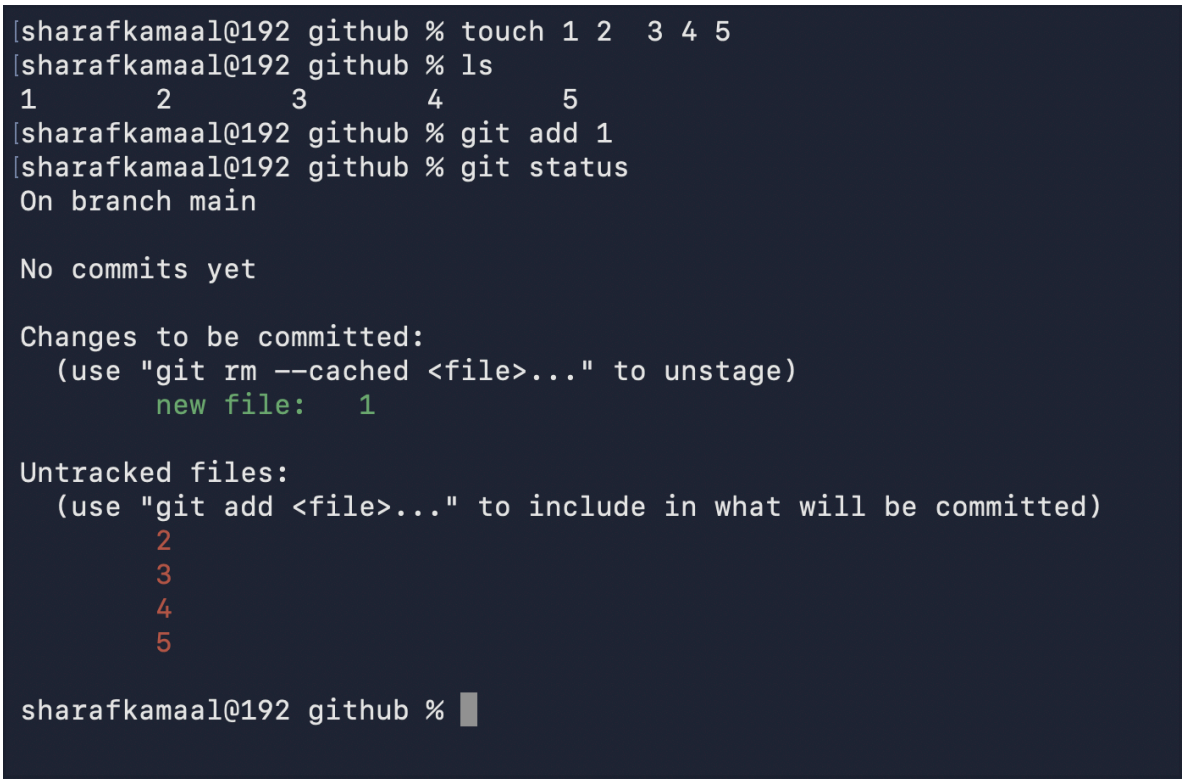
3. git --version – Displays the installed Git version.

```
sharafkamaal@192 github % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
safe.directory=/Users/sharafkamaal/dyad-apps/*
user.name=Sharaf Kamaal
user.email=sharafkamaal@gmail.com
alias.pushall=!git add . && git commit -m "update" && git push
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
sharafkamaal@192 github %
```

git config --list – Displays all Git configuration settings, such as username, email, and other configured options.

```
sharafkamaal@192 github % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
safe.directory=/Users/sharafkamaal/dyad-apps/*
user.name=Sharaf Kamaal
user.email=sharafkamaal@gmail.com
alias.pushall=!git add . && git commit -m "update" && git push
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
sharafkamaal@192 github % git config --global user.name "sharaf"
sharafkamaal@192 github % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
safe.directory=/Users/sharafkamaal/dyad-apps/*
user.name=sharaf
user.email=sharafkamaal@gmail.com
alias.pushall=!git add . && git commit -m "update" && git push
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
core.precomposeunicode=true
sharafkamaal@192 github % git config --global user.email "sharafkamaal@gmail.com"
sharafkamaal@192 github % git config --list
credential.helper=osxkeychain
init.defaultbranch=main
safe.directory=/Users/sharafkamaal/dyad-apps/*
user.name=sharaf
user.email=sharafkamaal@gmail.com
```

git config --global user.name "sharaf" – Sets your Git username globally for all repositories on your computer.
git config --list – Displays all Git configuration settings.
git config --global user.email "sharafkamaal@gmail.com" – Sets your Git email address globally for all repositories on your computer.
git config --list – Verifies that your Git username, email, and other configuration settings are saved.



1. touch 1 2 3 4 5 – Creates five empty files named 1, 2, 3, 4, and 5.

2. ls – Lists the files and directories in the current directory.

3. git add 1 – Stages file 1 so it can be included in the next commit.

4. git status – Shows the repository status, including staged and unstaged changes.

```
[sharafkamaal@192 github % git add .
[sharafkamaal@192 github % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   1
        new file:   2
        new file:   3
        new file:   4
        new file:   5

sharafkamaal@192 github %
```

git add . – Stages all new and modified files in the current directory and its subdirectories for the next commit.

```
[sharafkamaal@192 github % git commit -m "adding multiple files"
[main (root-commit) c3b23d3] adding multiple files
 5 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 1
 create mode 100644 2
 create mode 100644 3
 create mode 100644 4
 create mode 100644 5
sharafkamaal@192 github %
```

git commit -m "adding multiple files" – Saves the staged changes in the Git repository with the commit message "adding multiple files".

```
[sharafkamaal@192 github % git log
commit c3b23d343f79f43486cd7160c1f49c7b947dd9e8 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Thu Oct 22 14:49:27 2026 +0530

    adding multiple files
sharafkamaal@192 github %
```

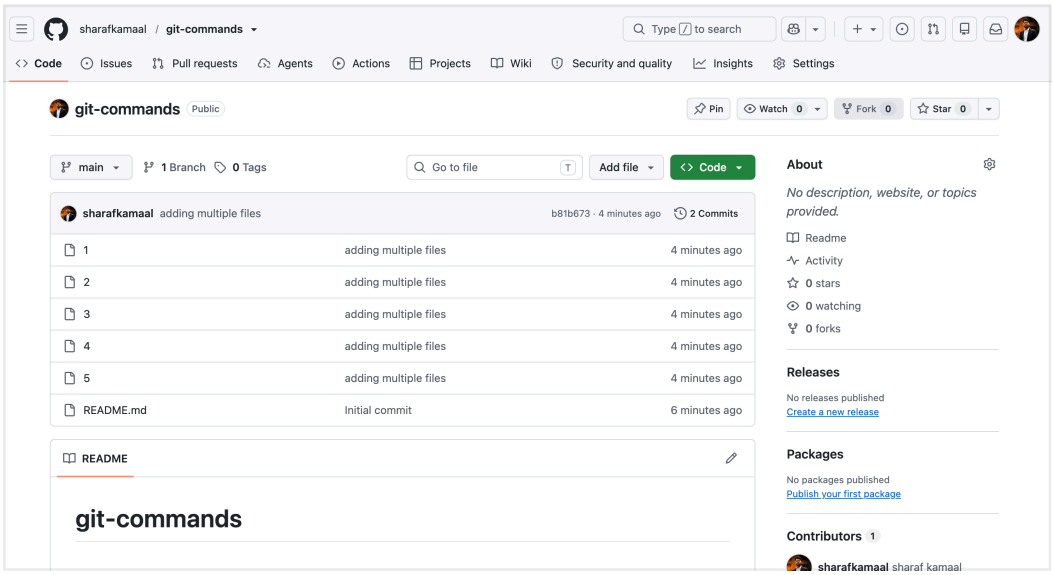
git log – Displays the commit history of the Git repository, including commit IDs, authors, dates, and commit messages.

```
and the repository exists.
sharafkamaal@192 github % git remote add origin https://github.com/sharafkamaal/git-commands.git
sharafkamaal@192 github % git remote -v
origin  https://github.com/sharafkamaal/git-commands.git (fetch)
origin  https://github.com/sharafkamaal/git-commands.git (push)
```

1. git remote add origin https://github.com/sharafkamaal/git-commands.git – Connects your local Git repository to the GitHub repository and names the remote origin.
2. git remote -v – Displays the configured remote repository URLs for fetching and pushing code.

```
sharafkamaal@192 github % git status
On branch main
nothing to commit, working tree clean
sharafkamaal@192 github % git pull --rebase origin main
From https://github.com/sharafkamaal/git-commands
 * branch            main       -> FETCH_HEAD
Successfully rebased and updated refs/heads/main.
sharafkamaal@192 github % git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 288.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sharafkamaal/git-commands.git
   1e0d748..b81b673  main -> main
branch 'main' set up to track 'origin/main'.
sharafkamaal@192 github %
```

1. git push –u origin main – Uploads your local main branch to GitHub and sets origin/main as its upstream branch for future pushes and pulls.
2. git pull --rebase origin main – Downloads the latest changes from GitHub's main branch and rebases your local commits on top of them.




```
sharafkamaal@192 github % git clone https://github.com/sharafkamaal/git-commands.git
Cloning into 'git-commands'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
sharafkamaal@192 github % ls
1               2               3               4               5               git-commands    README.md
sharafkamaal@192 github %
```

git clone https://github.com/sharafkamaal/git-commands.git — Downloads a copy of the GitHub repository to your local computer, including its files and commit history.

```
sharafkamaal@192 github % ls
1               2               3               4               5               git-commands    README.md
sharafkamaal@192 github % git pull
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 962 bytes | 320.00 KiB/s, done.
From https://github.com/sharafkamaal/git-commands
   b81b673..3b7a750  main       -> origin/main
Updating b81b673..3b7a750
Fast-forward
 what | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 what
sharafkamaal@192 github % ls
1               2               3               4               5               git-commands    README.md       what
sharafkamaal@192 github %
```

git pull — Downloads the latest changes from the remote repository and integrates them into your current local branch.

```
sharafkamaal@192 github % git add .
sharafkamaal@192 github % git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   34

sharafkamaal@192 github % git reset
sharafkamaal@192 github % git status
On branch main
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        34

nothing added to commit but untracked files present (use "git add" to track)
sharafkamaal@192 github %
```

git reset — Moves the current branch’s staging state or commit position to a specified point, depending on the options used.